

Lesson 1 — Sound and hearing

auditus · 29 September 2026 · test sounds in sounds/, made by make_sounds.py

Sound

A **longitudinal** pressure wave: compressions and rarefactions of the air travelling at about 343 m/s (20 °C). The air oscillates in place and passes the disturbance on.

Physical quantity	Perception	Units and notes
Frequency	Pitch	Hz; the octave is a 2:1 ratio (110 → 220 → 440 Hz)
Amplitude (pressure)	Loudness	dB, a logarithmic scale: +10 dB ≈ twice as loud
Spectrum (partials and their amplitudes)	Timbre	harmonic series: f, 2f, 3f...; the spectral envelope tells instruments apart

The ear

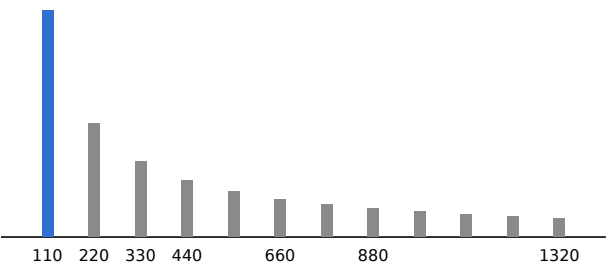
Part	Function
Outer (pinna, ear canal, eardrum)	collects the wave and turns it into vibration of the eardrum
Middle (malleus, incus, stapes)	impedance matching from air to cochlear fluid
Inner (cochlea)	tonotopy of the basilar membrane: high frequencies at the base, low at the apex; hair cells transduce into nerve signals. A biological spectrum analyser

Four properties of hearing

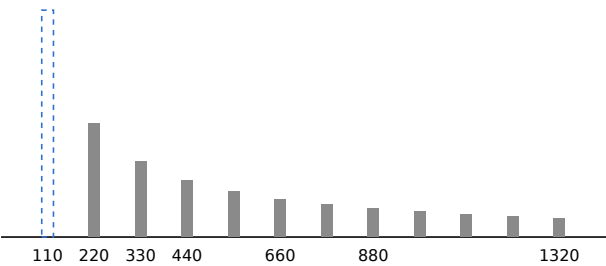
Phenomenon	In short
Hearing range and presbycusis	about 20 Hz–20 kHz when young; the upper limit drops with age (ISO 7029), mostly above 8 kHz
Equal-loudness contours (Fletcher–Munson, ISO 226)	highest sensitivity between 2 and 5 kHz; at low levels the threshold below 200 Hz rises steeply: bass "disappears" first
Binaural localisation	ITD (interaural time difference) below about 1.5 kHz, ILD (level difference) above; two equal speakers give a central phantom image
Residue pitch (missing fundamental)	the auditory system derives pitch from the periodicity of the harmonics, even without the fundamental: same chroma , less certain <i>pitch height</i>

The missing fundamental, in a picture

Complete A: fundamental 110 Hz + harmonics



Without fundamental: still heard as an A



On the right the 110 Hz bar is missing, yet the harmonics are still 110 Hz apart: the perceived pitch is still an A. This is why the low E at 82 Hz is recognisable even on speakers that reproduce little at 82 Hz.

Results of 29 September

Test (file)	Result	Reading
Sweep 20 Hz–16 kHz (04)	heard from 4.0 s to 19.26 s: about 76 Hz - 12.5 kHz	low limit = speakers + level; high limit at or above average at 60 ($\pm 10\%$ for the hand-held stopwatch)
Pure tone / A with harmonics (01, 03)	same note, the pure tone sounds higher	timbre mistaken for pitch; the speakers attenuate 110 Hz
Left, right, centre (05)	phantom image in the centre	balanced speakers, symmetrical listening position
Missing fundamental (06)	same note, different timbre, uncertain register	residue pitch: same chroma
Listening level	OUTPUT at 12 o'clock instead of 9	sines peaking at -20 dBFS (about -23 dBFS RMS), 10 dB below mastered music

Sweep formula

Logarithmic sweep from 20 to 16 000 Hz in 20 s: frequency at second t

$$f(t) = 20 \cdot 800^{(t/20)} \text{ Hz} \quad t = 4 \text{ s} \rightarrow 76 \text{ Hz} \quad t = 19.26 \text{ s} \rightarrow 12.5 \text{ kHz}$$

To listen again

```
cd ~/Documents/projects/auditus/lessons/01_sound_and_hearing
for f in sounds/*.wav; do echo "$f"; aplay -q "$f"; done
```

Before going back to music, turn OUTPUT back to 9 o'clock.